



Thanh Vuong Dang

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Hồ Chí Minh

R&D Engineer

Passionate about applied mathematics, physics and computer sciences. Seeking for a job in technology company.

Working experience

Algorithm researcher
04/2026 - current

Viettel High Tech

- Develop optimization algorithm for sensing position problem in 5G NR.
- Develop simulation algorithm for designing systolic matrix architecture of CPU.
- GPU programming for designing GPU architecture on FPGA chip.
- Link: <https://viettelhightech.vn/>

Researcher

03/2025 - 06/2025

Simidea R&D

- Develop Jpythmen library (Java) for Johnson–Champoux–Allard model and Biot model in impedance tube.
- Simulate the physical experiment using Groovy console, Java, Symplegma Development Environment (SDE) and COMSOL to compute absorption, reflexion, transmission and dissipation coefficient of solid materials.
- Implement inverse characterization to identify Lamé coefficients and density of solid materials based on real experiment data conducted at company utilizing Differential Evolutionary Algorithm.
- Link: <https://simidea.es/>

Academic research position

04/2019–12/2023

University Paris 13, Paris, France

- Solve optimization problems with internal wave constraints by Finite Volume Method with MATLAB, Python and C/C++ on Ubuntu.
- Solve optimal control problems of wave equation with boundary control data by Finite Difference Method with MATLAB, Python and C/C++ on Ubuntu.

Education

PhD in Applied Mathematics
10/2020–12/2023

University Paris 13, Paris, France

- Thesis: Parallel in time algorithms for internal wave controlled problem.
- Supervisors: Professor Laurence Halpern and Dr. Bérangère Delourme.
- Link thesis: <https://hal.science/tel-04701863>.

Master in Applied Mathematics
2019–2020

University of Orléans, Orléans, France

- Thesis: Schwarz waveform relaxation algorithms for optimal control problem constrained by the wave equation, GPA: 17/20.
- Supervisors: Professor Laurence Halpern and Dr. Bérangère Delourme.

Bachelor in Mathematics
2015–2019

University of Sciences, Ho Chi Minh, Viet Nam

- Thesis: Morrey spaces, GPA: 9.16/10.
- Supervisors: Professor Duong Minh Duc.
- Studying Harmonic Analysis with mathematical theorems and inequalities.

Scholarship

2016–2019

National scholarships to develop mathematics from 2010–2020

Vietnam Institute for Advanced Study in Mathematics (VIASM)

Publication

On the boundary controlled wave equation

- Author: Thanh Vuong Dang
- Link: <https://hal.science/hal-04463449>

Time parallelization of a distributed optimal control problem for the wave equation.

- Author: Thanh Vuong Dang, Laurence Halpern, Bérangère Delourme
- 28th International Conference on Domain Decomposition Methods (DD28)

Technical Skill

System
Software
Program
Database

Languages		
Windows, Linux (Unbuntu)	Vietnamese	Mother tongues
LaTeX, Maple, COMSOL	French	BI
C/C++, Java, Python, MATLAB, Groovy	English	CI
SQL, Pandas		

References

Laurence Halpern
Professor Emeritus / Paris 13 Univeristy
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Luis Sanchez Ricart
Professor / University of Cartagena
Founder at Simidea R&D
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MCF / Paris 13 University
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